Question 1: See this example dialog with D`ukeGPT asking the LLM to explain something about static methods. Choose a Java concept that you have a question about, ask DukeGPT about the concept, and download the dialog as a PDF which you'll include as part of the PDF you upload to Gradescope.

Question 2: What is the smallest value of variable `threshold` in `Person201Nearby` that yields exactly three people near to Ricardo. Write a few sentences about how you found the value, essentially providing an algorithm for anyone to replicate your work so that with a different/new data file they could find the smallest such value **efficiently**.

The lowest is 17.

Start with the threshold high.
Decrease the threshold by 1 until nearby persons reaches 2.
Add one.

e.g. 50 -> 17 is three people. At 16 is two people. 16+1 = 17, thus 17 is the threshold lowest for 3 people.

Question 3: The online data for this project read via URL is **not encrypted**. See this DukeGPT dialog about why it might be a good idea to encrypt and provide a few sentences about what makes sense and what doesn't in the answer DukeGPT provided.

Encryption in URL allows for more secure transfer of data between the server and project. This prevents unwanted traffic or malicious activity on the data.

Question: what does the historical development of encrypting data look like?

Question 4: According to the `.equals` method of the `Person201` class, when are two `Person201` objects considered to be equal? Is it case sensitive for their names or for their phrases? (Case sensitive means different answers are returned depending on capitalization). If you use DukeGPT or an LLM to help answer this question, please include that dialog.

They have the same name instance variable value, the same longitude, same latitude, and same eatery value.

It is case sensitive as it utilizes the String equals method, which requires the strings to be identical.

Question 5: Copy/paste the output you get from modifying the method `countEateries` in the file `CountEateries.java` based on the description above.

```
Information for 32 different eateries
0:     Alpaca
0:     Banh's Cuisine
0:     Chicken Bee
0:     Common Market
0:     Cosmic Cantina
0:     Dain's Place
0:     Devil's Pizzeria
0:     Elmo's Diner
0:     Genji
0:     Guasaca
0:     Happy + Hale
0:     Heavenly Buffaloes
0:     International Delights
0:     Jimmy John's
0:     Juju
0:     Kabab and Curry
0:     Kiichi Ramen
0:     Lime and Lemon
0:     Local 22
0:     Mad Hatter Cafe
0:     Metro 8 Steakhouse
0:     Mi Cavillo Antojitos Mexican
0:     Monuts
0:     Panera
0:     Parizade
0:     Playa Bowls
0:     Shuckin' Shack
0:     Subway
0:     The Loop
0:     The Tavern
0:     Vin Rouge
0:     Zenfish Poke Bar
--------
total = 0
total # eateries = 32
PS C:\Users\Jason\Documents\CS201\Projects\p0-person-201> ^C
PS C:\Users\Jason\Documents\CS201\Projects\p0-person-201>
PS C:\Users\Jason\Documents\CS201\Projects\p0-person-201>  c:; cd
'c:\Users\Jason\Documents\CS201\Projects\p0-person-201'; & 'C:\Program Files\Eclipse
Adoptium\jdk-17.0.20.101-hotspot\bin\java.exe'
```

'@C:\Users\Jason\AppData\Local\Temp\cp_70qod7dlv530mqg2b8yebjubd.argfile'
'CountEateries'
Information for 32 different eateries
4: Alpaca
3: Banh's Cuisine
4: Chicken Bee
3: Common Market
2: Cosmic Cantina
5: Dain's Place
3: Devil's Pizzeria
5: Elmo's Diner
4: Genji
1: Guasaca
1: Happy + Hale
2: Heavenly Buffaloes
2: International Delights
2: Jimmy John's
1: Juju
1: Kabab and Curry
4: Kiichi Ramen
4: Lime and Lemon
5: Local 22
3: Mad Hatter Cafe
4: Metro 8 Steakhouse
4: Mi Cavillo Antojitos Mexican
1: Monuts
3: Panera
3: Parizade
4: Playa Bowls
1: Shuckin' Shack
3: Subway
3: The Loop
2: The Tavern
6: Vin Rouge
4: Zenfish Poke Bar
---
total = 97
total # eateries = 32

Question 6: Explain how, if at all, you used LLMs in completing this assignment.

LLM's were used as per the project instructions to explain a specific Java concept, as well as explain more regarding data encryption.
Otherwise no LLM usage was employed.

Challenge: If you complete the challenge, please write up what you found.

First, the same technique to retrieve data information and bundle into an array of Person201 objects was utilized from PeopleDownloader. Then a similar technique of finding the occurrence of each eatery was recorded, and bundled into a Map which paired each eatery with its occurrence. A while loop is used to iteratively find the max occurrence in the map and then removing it in order to rank each eatery. This yields big O(n^2) efficiency.

The following info is found from the URL:
rank: 1: occurences: 78: Cosmic Cantina
rank: 2: occurences: 71: Alpaca
rank: 3: occurences: 42: Juju
rank: 4: occurences: 40: Szechuan Mansion Hotpot
rank: 5: occurences: 34: Subway
rank: 6: occurences: 32: Panera
rank: 7: occurences: 24: Chipotle
rank: 8: occurences: 20: Wheat
rank: 9: occurences: 14: Playa Bowls
rank: 10: occurences: 12: Kiichi Ramen
rank: 11: occurences: 10: Elmo's Diner
rank: 12: occurences: 10: Chicken Bee
rank: 13: occurences: 10: Mad Hatter Cafe
rank: 14: occurences: 7: Lime & Lemon
rank: 15: occurences: 7: Monuts
rank: 16: occurences: 6: Banh's Cuisine
rank: 17: occurences: 6: Lime and Lemon
rank: 18: occurences: 5: Vin Rouge
rank: 19: occurences: 5: Happy + Hale
rank: 20: occurences: 4: Guasaca
rank: 21: occurences: 3: Tous Les Jours
rank: 22: occurences: 3: Jimmy John's
rank: 23: occurences: 3: Dain's Place
rank: 24: occurences: 2: Local 22
rank: 25: occurences: 2: The Loop
rank: 26: occurences: 2: Koi Sushi and Hibachi
rank: 27: occurences: 2: Kabab and Curry
rank: 28: occurences: 2: International Delights
rank: 29: occurences: 2: Metro 8 Steakhouse
rank: 30: occurences: 2: Heavenly Buffaloes
rank: 31: occurences: 1: Devil's Pizzeria
rank: 32: occurences: 1: Common Market
rank: 33: occurences: 1: The Tavern
rank: 34: occurences: 1: Zenfish Poke Bar

rank: 35: occurences: 1: Parizade
rank: 36: occurences: 1: Genji
total = 466